

Bidut Sharkar Shemanto

Backend Focused Full-Stack Engineer — AI Integration & DevOps

+8801521721778 — ✉ shemantosharkarofficial@gmail.com — in /in/shemanto/ — /shemanto27
 📍 Dhaka, Bangladesh

Summary — Backend-focused Full-Stack Developer with experience in Web and App development using **Django REST Framework, React.js and React Native**. Skilled in AI integration and building and deploying applications with documented APIs, responsive UIs, and cloud deployment using **AWS, Docker, and GitHub Actions**.

Education and Certifications

Khulna University of Engineering and Technology (KUET)
B.Sc. Eng. in Energy Science and Engineering

2020 – 2025

Certifications

- Data Science Certifications (from Ostad Platform) - Python, Machine Learning, Data Analysis, SQL
- Web Development (from Programming Hero Platform) - HTML, CSS, JavaScript, React.js, Express.js, MongoDB
- 90-Hour Machine Learning with Python Training (from ICT Division under EDGE Program at KUET)

Skills and Tools

Languages: Python, JavaScript/TypeScript, Go

Frontend: React.js, React Native, HTML, CSS, Tailwind CSS

Backend: Django, FastAPI, Node.js, Express.js, Redis, Celery, Websocket, RabbitMQ

Databases: PostgreSQL, MongoDB, Vector Databases (ChromaDB, Pinecone)

Machine Learning & AI: Pandas, NumPy, Scikit-learn, PyTorch, LangChain, Vector Databases, RAG, MCP

DevOps & Cloud: Git, AWS, Linux (Ubuntu), Nginx, Docker, CI/CD (Github Action), Terraform, Ansible

Experience

Sparktech Agency
Full Time

Aug 2025 – Present (10 Months)



Full-Stack Developer (Promoted)

Feb 2026 – Present (4 Months)

- **Promoted to Full-Stack Developer** after demonstrating strong backend, frontend and DevOps skills. Woked with React.js for frontend and deployed backend and frontend in AWS with proper DevOps workflow.



Python Developer

Aug 2025 – Feb 2026 (6 Months)

- **Worked in the AI team as Backend Developer;** created REST APIs for Web and Flutter Mobile App, integrated AI features using OpenAI and LangChain.
- **Delivered 6+ projects with 5-star client ratings;** assisted team leader with requirement checks, client communication, budget negotiation, and quotation making.

Projects

MenuSideKick - AI powered Flutter App for Scanning Restaurant Menu (*Company Project)

• **Project Video** →

Django REST Framework, Flutter Mobile App, AWS RDS, AWS Textract OCR, AI ChatBot(OpenAI) Docker, AWS, GitHub Actions

- Collaborated closely with the Flutter team and AI team to build an AI-powered restaurant menu-scanning app, ensuring smooth communication between mobile, backend, and AI components.
- Integrated the AI team's **OCR and chatbot** modules into the Django backend, enabling automated menu extraction and conversational food-recommendation features.
- Containerized the backend using Docker and deployed it with Nginx on AWS EC2; implemented CI/CD pipelines with GitHub Actions for automated builds and deployments.

WonderTales — AI-Powered Personalized Bedtime Story App (*Company Project)

• **Live Site** →

Django REST Framework, Flutter Mobile App, ElevenLabs, OpenAI, AWS, Docker, CI/CD

- **Integrated ElevenLabs** voice synthesis API to enable high-quality AI narration and **parent voice cloning**, allowing stories to be told in a child's own family voice.
- Built and customized the **Django backend and admin panel**; developed REST APIs consumed by the Flutter mobile app for story generation, user profiles, and subscription management.

- Developed the **public landing page** (wondertaleshub.com) with a blog system; handled full **DevOps** — Dockerized services, AWS EC2 deployment, Nginx, and CI/CD pipelines.

Mon5Monjeur - NBR Fantasy Game (*Company Project)

•Project Video →

Django REST Framework, Redis, Websocket, Flutter App, React.js

- Worked closely with the Flutter team to build a real-time fantasy gaming app backend, collaborating on gameplay flow, scoring logic, and API contracts.
- Implemented complex game mechanics and leaderboard logic in Django REST Framework, integrating external sports APIs to deliver live player and match data.
- Built WebSocket-based real-time communication for live matches, pagination, enabling instant score updates and in-game interactions between users.

GamePrice AI - AI Game Box Cover Analysis to Give Price (*Company Project)

•Project Video →

Next.js, Django, Docker, External API Integration, AI Integration

- **Integrated the AI team's model into the backend** to analyze game box cover images and estimate game prices using data pulled from the **PriceCharting external API**.
- Developed and maintained Django-based REST APIs to handle image uploads, AI inference requests, and real-time price lookup workflows.
- **Assisted the frontend developer in Dockerizing the Next.js application** and resolving deployment issues, ensuring the full-stack system could be deployed smoothly on AWS.

ScholarPage — Full-Stack Academic Portfolio Builder (*Personal Project)

•Project Video →

React.js, Django REST, PostgreSQL (Neon), JWT Auth, TailwindCSS, Docker, AWS, GitHub Actions

- Built a full-stack web application that enables researchers to create interactive academic portfolios, including education, publications, and research interests.
- Developed secure REST APIs using Django and PostgreSQL; built a responsive frontend with React.js and Tailwind CSS.
- Deployed on AWS EC2 using Docker; implemented JWT authentication and set up CI/CD with GitHub Actions.
- Configured Nginx and Gunicorn to serve the backend in a production-ready setup on Ubuntu-based EC2 instances.

PeoplesList — Member Directory Platform for Groups and Communities (*Personal Project)

•Project Video →

Django REST Framework, PostgreSQL, React.js, Docker, AWS EC2, Nginx

- Built a tool for managing member directories across groups (e.g., alumni, clubs) without requiring member sign-ups, Combining Google Form + Sheet in single page
- Used just one link to add and view members, common password, and moderation options.
- Utilized DRF and PostgreSQL for scalable API backend; deployed via Docker on AWS EC2 with Nginx.

Achievements

- **Champion in Inter-University Innovation Idea and Project Competition-2022** (Issued by FabLab KUET, Dec 2022)
- **3rd Place in All Bangladesh Asteroid Search Campaign** (Issued by NASA, Oct 2019)
- **Finalist in Nuclear Dilemma 6** (Issued by Information Center on Nuclear Energy, Apr 2023)
- **Finalist in AGRIBIZ, Agriculture-Based Inter-University Business Case Competition** (Issued by Sher-e-Bangla Agriculture University, Jan 2023)
- **4th Place in IEOM Society KUET Student Chapter's Innovative Idea Contest** (Issued by IEOM Society KUET Student Chapter, Jul 2022)
- **1st Runner-Up in BizBash 2.0** (Issued by KUET Career Club-KCC)

Extracurricular Activities

KUET Business and Entrepreneurship Club

Mar 2022 – Sep 2023

Senior Executive: Public Relations (PR) and Graphics Design

- Organized **TEDx KUET** as part of the graphics and content team.
- Conducted the **CASE CRACK WORKSHOP 2022** as a Junior Executive.
- Organized and managed the **HULT Prize at KUET**; conducted workshops on presentation skills, creating PowerPoint slides, and business idea showcasing.

KRIoTIC (KUET Robotics and IoT Innovation Club)

May 2022 – Mar 2023

General Member

- Organized and conducted workshops on **IoT and microcontrollers**.
- Delivered a **speech on IoT at Khulna University to 200+ people**, raising awareness about IoT and frontier technologies.